

# Discrete-Time Signals

May 13, 2026

# What is a Discrete-Time Signal?

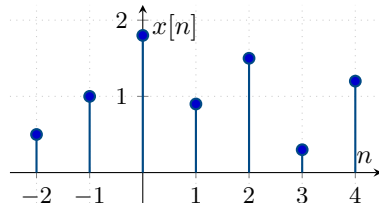
## Definition

A **discrete-time signal**  $x[n]$  is a function defined only at integer values of  $n \in \mathbb{Z}$ , i.e. a sequence of numbers:

$$x[n] = \{\dots, x[-1], x[0], x[1], x[2], \dots\}$$

- $n$  is the **sample index**
- Signal is *undefined* between integer values of  $n$
- Represented as a **stem plot**

Generic Discrete-Time Signal



# Discrete-Time Signals

## Discrete-Time Signal

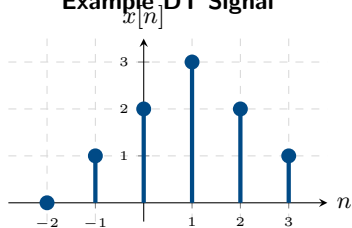
A **DT signal**  $x[n]$  is a function defined only at *integer* values  $n \in \mathbb{Z}$ . Between integers the signal is **undefined**.

Three equivalent representations:

- 1 **Functional** — closed-form mathematical expression
- 2 **Tabular** — table listing  $n$  vs.  $x[n]$
- 3 **Sequence** — ordered list with origin marker  $\uparrow$

$$x : \mathbb{Z} \rightarrow \mathbb{R}, \quad x[n] \text{ defined } \forall n \in \mathbb{Z}$$

## Example DT Signal



*Samples exist only at integer  $n$ ; the dots are the signal.*

# 1 — Functional Representation

## Standard building blocks:

$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases} \quad u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

## Derived signals:

- Real exponential:  $x[n] = A a^n u[n]$
- Sinusoidal:  $x[n] = A \cos(\omega_0 n + \phi)$
- Rectangular:  $x[n] = u[n] - u[n - N]$

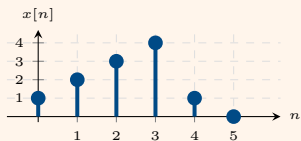
## Key Advantage

One formula describes *infinitely many* samples compactly.

# Functional Representation — Further Examples

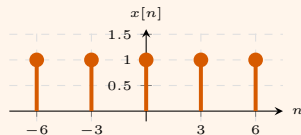
## Piecewise Signal

$$x[n] = \begin{cases} n + 1, & 0 \leq n \leq 3 \\ 5 - n, & 4 \leq n \leq 5 \\ 0, & \text{otherwise} \end{cases}$$



## Periodic Impulse Train

$$x[n] = \sum_{k=-\infty}^{\infty} \delta[n - 3k], \quad N = 3$$



## 2 — Tabular Representation

### Definition

List sample values in a **two-row table**: one row for the index  $n$ , one row for the amplitude  $x[n]$ .

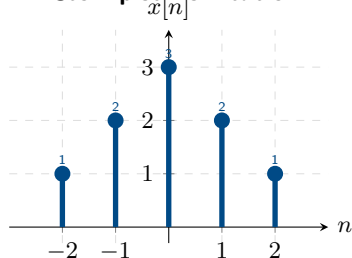
### When to use:

- Finite-length or short signals
- Verifying computed values
- Simulation / numerical work

### Example: triangular pulse

| $n$    | -2 | -1 | 0 | 1 | 2 |
|--------|----|----|---|---|---|
| $x[n]$ | 1  | 2  | 3 | 2 | 1 |

Stem plot from table



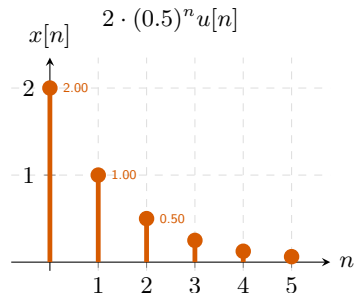
# Tabular Representation — Example

**Signal:**  $x[n] = 2 \cdot (0.5)^n u[n]$

**Table of values**

| $n$    | 0 | 1 | 2    | 3    | 4    | 5    | ... |
|--------|---|---|------|------|------|------|-----|
| $x[n]$ | 2 | 1 | 0.50 | 0.25 | 0.13 | 0.06 | ... |

- Table shows a **finite window** of an infinite signal



### 3 — Sequence Representation

#### Definition

Write  $x[n]$  as an ordered list of values. The **time origin** ( $n = 0$ ) is marked with an upward arrow  $\uparrow$  placed below the corresponding sample.

#### General notation:

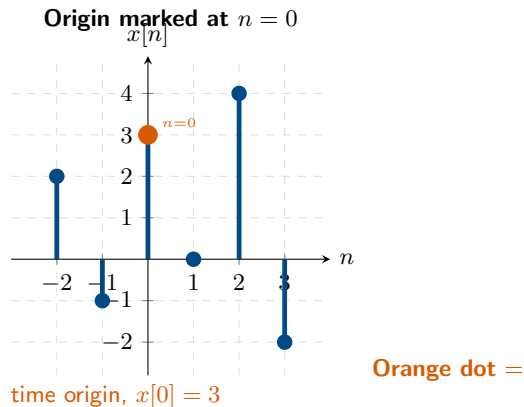
$$x[n] = \{\dots, x_{-2}, x_{-1}, x_0, x_1, x_2, \dots\}$$

$\uparrow$

#### Example

$$x[n] = \{2, -1, 3, 0, 4, -2\}$$

$\uparrow$





# Sequence Representation — Origin Position Matters

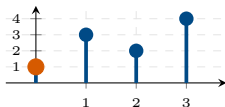
## Key Rule

The *same values* yield **different signals** depending on where the  $\uparrow$  is placed. Always state the origin explicitly.

**Case A** — origin at 1st sample:

$$x_A[n] = \{1, 3, 2, 4\}$$

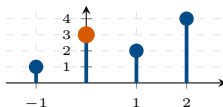
$\uparrow$



**Case B** — origin at 2nd sample:

$$x_B[n] = \{1, 3, 2, 4\}$$

$\uparrow$



## Relationship

Shifting the origin = **time shift** of the signal.

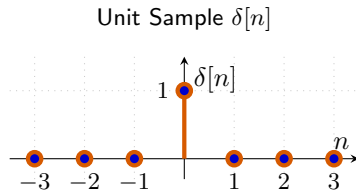
$$x_B[n] = x_A[n + 1]$$

## Some Elementary Discrete-Time Signals

# Unit Sample Sequence $\delta[n]$

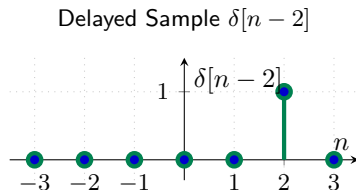
## Definition

$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$



## Delayed Impulse

$$\delta[n - k] = \begin{cases} 1, & n = k \\ 0, & n \neq k \end{cases}$$



- Nonzero only at one sample
- Used to represent discrete-time signals
- Also called the **discrete impulse**

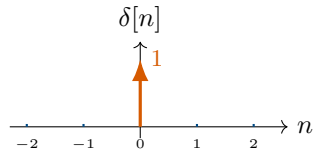
# What is the Unit Impulse (Delta) Signal?

## Definition: Unit Impulse $\delta[n]$

$$\delta[n] = \begin{cases} 1 & n = 0 \\ 0 & \text{otherwise} \end{cases}$$

## Key Properties:

- Single non-zero value at  $n = 0$
- Also called a “spike” or Dirac-like sequence
- The 0<sup>th</sup> sample equals 1



Shifted impulse:  $\delta[n - k]$  is 1 only at  $n = k$ , zero elsewhere.

# Scaled & Shifted Impulse

**Notation:**  $m \delta[n + k]$

- $m$  — **scale** (amplitude multiplier)
- $k$  — **shift** (time index where spike occurs)

$$m \delta[n + k] = m \quad \text{at } n = -k, \quad 0 \text{ otherwise}$$

**Examples:**

$$3 \delta[n - 2] \Rightarrow \text{spike of height 3 at } n = 2$$

$$-2 \delta[n + 1] \Rightarrow \text{spike of height } -2 \text{ at } n = -1$$

# Time Shifting

## Delay (Right Shift)

$$y[n] = x[n - k], \quad k > 0$$

Signal moves **right** by  $k$  samples.  
(Introduces time delay)

## Advance (Left Shift)

$$y[n] = x[n + k], \quad k > 0$$

Signal moves **left** by  $k$  samples.  
(Advances in time)

## Quick Check

If  $x[n] = \{1, 2, 3\}$  starting at  $n = 0$ , then  
 $x[n - 2] = \{1, 2, 3\}$  starting at  $n = 2$ .

# Every Signal as a Sum of Impulses

## Core Theorem

*“Every signal is a summation of a bunch of **shifted & scaled** Unit Impulses.”*

**Formal expression:**

$$x[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n - k]$$

Each term  $x[k] \delta[n - k]$  is a scaled, shifted impulse that “picks out” the value  $x[k]$  at time  $k$ .

# Unit Sample – Worked Example

## Example 1

Express the signal  $x[n] = \{2, 0, -1, 3\}$  starting at  $n = -1$  as a sum of scaled impulses.



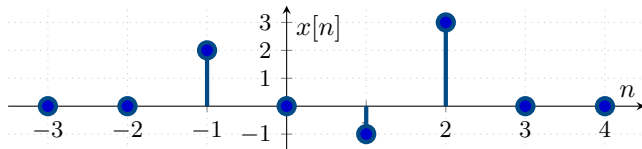
# Unit Sample – Worked Example

## Solution

The non-zero samples are at  $n = -1, 0, 1, 2$  with values 2, 0, -1, 3.

$$x[n] = 2\delta[n+1] + 0 \cdot \delta[n] + (-1)\delta[n-1] + 3\delta[n-2]$$

$$x[n] = 2\delta[n+1] - \delta[n-1] + 3\delta[n-2]$$



# Step-by-Step Decomposition Example

Given signal with origin at  $n = 0$ :

$$x[n] = \{2, \underset{\uparrow}{4}, 0, 3\}$$

# Step-by-Step Decomposition Example

Given signal with origin at  $n = 0$ :

$$x[n] = \{2, \underset{\uparrow}{4}, 0, 3\}$$

**Step 1:** Extract each sample as a scaled, shifted impulse:

$$\{2, 0, 0, 0\} = 2\delta[n + 1]$$

$$\{0, 4, 0, 0\} = 4\delta[n]$$

$$\{0, 0, 0, 0\} = 0\delta[n - 1]$$

$$\{0, 0, 0, 3\} = 3\delta[n - 2]$$

# Step-by-Step Decomposition Example

Given signal with origin at  $n = 0$ :

$$x[n] = \{2, \underset{\uparrow}{4}, 0, 3\}$$

**Step 1:** Extract each sample as a scaled, shifted impulse:

$$\{2, 0, 0, 0\} = 2\delta[n + 1]$$

$$\{0, 4, 0, 0\} = 4\delta[n]$$

$$\{0, 0, 0, 0\} = 0\delta[n - 1]$$

$$\{0, 0, 0, 3\} = 3\delta[n - 2]$$

**Step 2:** Sum them up:

$$x[n] = 2\delta[n + 1] + 4\delta[n] + 0\delta[n - 1] + 3\delta[n - 2]$$

$\Rightarrow$  Verified:  $\{2, 4, 0, 3\}$  ✓

## Definition

Impulse decomposition breaks an  $N$  samples signal into  $N$  component signals, each containing  $N$  samples. Each of the component signals contains one point from the original signal, with the remainder of the values being zero. A single nonzero point in a string of zeros is called an impulse.

# Impulse Decomposition

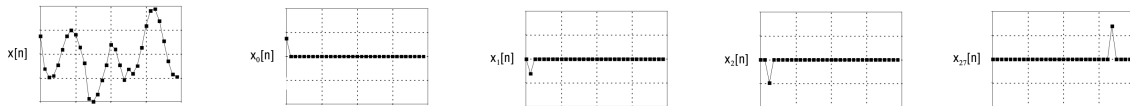


Figure: Impulse Decomposition

# Unit Step Sequence $u[n]$

## Definition

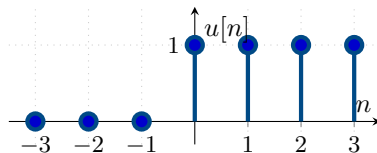
$$u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

## Relation to impulse:

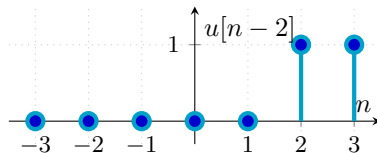
$$u[n] = \sum_{k=0}^{\infty} \delta[n-k] = \sum_{k=-\infty}^n \delta[k]$$

$$\delta[n] = u[n] - u[n-1]$$

Unit Step  $u[n]$



Delayed Step  $u[n-2]$



# Unit Step – Worked Example

## Example 2

Express the rectangular pulse  $x[n] = 1$  for  $2 \leq n \leq 5$ , zero elsewhere, using unit step functions.



# Unit Step – Worked Example

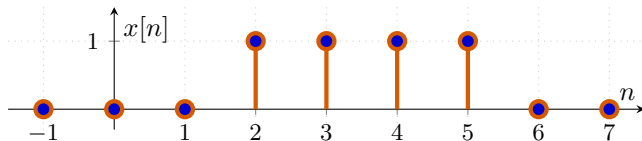
## Solution

A pulse that *starts* at  $n = 2$  is  $u[n - 2]$ .

A pulse that *ends* at  $n = 5$  means subtracting  $u[n - 6]$ :

$$x[n] = u[n - 2] - u[n - 6]$$

Verify: for  $n = 2, 3, 4, 5$ :  $1 - 0 = 1$ . For  $n \geq 6$ :  $1 - 1 = 0$ . ✓



# Unit Ramp Sequence $r[n]$

## Definition

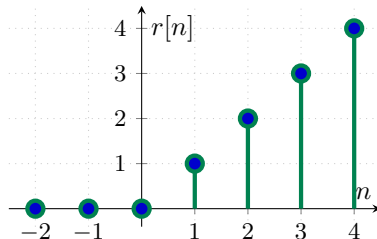
$$r[n] = \begin{cases} n, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

## Relations:

$$r[n] = \sum_{k=0}^{\infty} u[n-k] = \sum_{m=-\infty}^n u[m]$$

$$u[n] = r[n] - r[n-1]$$

Unit Ramp  $r[n] = n u[n]$



# Exponential Sequence $x[n] = A a^n$

## Definition

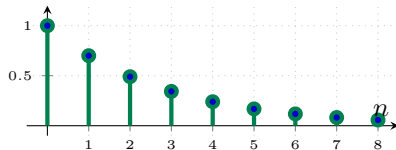
$$x[n] = A a^n, \quad n \in \mathbb{Z}$$

where  $A \in \mathbb{R}$  (amplitude) and  $a \in \mathbb{R}$  (base).

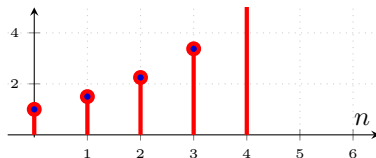
## Behaviour by $|a|$ :

- $|a| < 1$ : **decaying** envelope
- $|a| > 1$ : **growing** envelope
- $|a| = 1$ : **constant** magnitude ( $a = \pm 1$ )
- $a < 0$ : signal **alternates** in sign

Decaying:  $a = 0.7$



Growing:  $a = 1.5$



# Symmetric (Even) Signals

## Definition – Even Signal

A signal  $x[n]$  is **even** (symmetric) if

$$x[-n] = x[n] \quad \forall n \in \mathbb{Z}$$

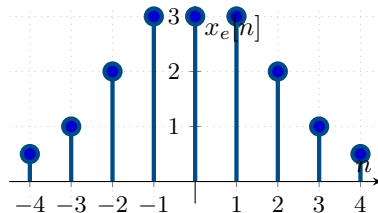
**Even part of any signal:**

$$x_e[n] = \frac{x[n] + x[-n]}{2}$$

**Properties:**

- Mirror symmetry about  $n = 0$
- $\sum_{n=-\infty}^{\infty} x_e[n]$  is always real (if  $x$  is real)
- $\cos(\omega_0 n)$  is even

Even Signal Example



# Anti-Symmetric (Odd) Signals

## Definition – Odd Signal

A signal  $x[n]$  is **odd** (anti-symmetric) if

$$x[-n] = -x[n] \quad \forall n \in \mathbb{Z}$$

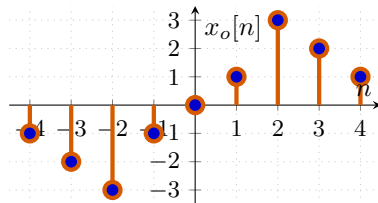
Odd part of any signal:

$$x_o[n] = \frac{x[n] - x[-n]}{2}$$

Properties:

- **Always**  $x_o[0] = 0$
- $\sin(\omega_0 n)$  is odd
- $\sum_{n=-\infty}^{\infty} x_o[n] = 0$

Odd Signal Example



# Even–Odd Decomposition Theorem

## Theorem

**Every** real signal  $x[n]$  can be *uniquely* decomposed as:

$$x[n] = x_e[n] + x_o[n]$$

where

$$x_e[n] = \frac{x[n] + x[-n]}{2} \quad (\text{even part}), \quad x_o[n] = \frac{x[n] - x[-n]}{2} \quad (\text{odd part})$$

## Quick checks:

- $x_e[-n] = +x_e[n]$  ✓ (even symmetry preserved)
- $x_o[-n] = -x_o[n]$  ✓ (odd symmetry preserved)
- $x_e[n] + x_o[n] = \frac{x[n] + x[-n]}{2} + \frac{x[n] - x[-n]}{2} = x[n]$  ✓

# Folding (Reflection)

## Time Reversal / Folding

$$y[n] = x[-n]$$

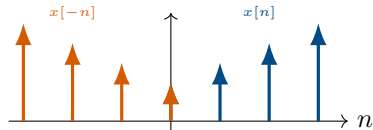
Reflects the signal about the **vertical axis** ( $n = 0$ ).

Also called *folding* — a **virtual** operation (not physical delay).

### Example:

$x[n] = \{1, 2, 3, 4\}$  at  $n = 0, 1, 2, 3$

$x[-n] = \{4, 3, 2, 1\}$  at  $n = -3, -2, -1, 0$



# Combined Transformation: $x[-n + 2]$

**Problem: Find  $x(-n + 2)$  from  $x[n]$**

Two valid interpretations — **Order Matters!**

## Method A (standard):

① Reflect:  $x[n] \rightarrow x[-n]$

② Shift left by 2:

$$x[-n] \rightarrow x[-(n - 2)] = x[-n + 2]$$

Net effect: **right shift by 2**

## Clarification:

$$x[-(n - 2)] = x[-n + 2]$$

Note:  $x[-(n - 2)]$  means replace  $n$  with  $n - 2$  *after* folding, which shifts right.

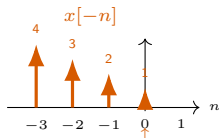


# Combined Transformation: $x[-n + 2]$

**Problem:** given  $x[n]$

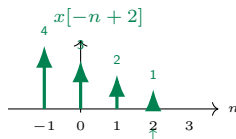
Two approaches — **Order Matters!**

**Step 1: Reflect**  $x[n] \rightarrow x[-n]$

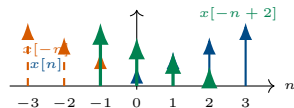


**Step 2: Shift right by 2**

$$x[-n] \rightarrow x[-(n - 2)] = x[-n + 2]$$



**All three overlaid**



# Down Sampling

## Definition: Down Sampling by factor $M$

$$y[n] = x[Mn]$$

Keep every  $M^{\text{th}}$  sample of  $x[n]$ .

**For  $M = 2$ :**  $y[n] = x[2n]$

|                    |     |        |        |        |        |        |
|--------------------|-----|--------|--------|--------|--------|--------|
| $n$ (original)     | ... | 0      | 1      | 2      | 3      | 4      |
| $x[n]$             | ... | $x[0]$ | $x[1]$ | $x[2]$ | $x[3]$ | $x[4]$ |
| $n$ (down-sampled) | ... | 0      |        | 1      |        | 2      |
| $y[n] = x[2n]$     | ... | $x[0]$ |        | $x[2]$ |        | $x[4]$ |

The 2<sup>nd</sup> sample of original  $\rightarrow$  1<sup>st</sup> sample of down-sampled signal.  
Operates on a *sample-to-sample* basis.

## Total Energy of a Discrete-Time Signal

$$E_N = \sum_{n=-\infty}^{\infty} |x[n]|^2$$

The  $|\cdot|^2$  handles **complex-valued** signals (takes magnitude squared).

**Real signals:**

$$E_N = \sum_n x[n]^2$$

**Complex signals** ( $x[n] = a + jb$ ):

$$|x[n]|^2 = a^2 + b^2$$

## Example

$$x[n] = \{2, 4, 0, 3\}$$

$$E_N = 2^2 + 4^2 + 0^2 + 3^2 = 4 + 16 + 0 + 9 = \mathbf{29}$$

# Periodic vs. Aperiodic Signals

A discrete-time signal  $x[n]$  is **periodic** with period  $N$  if:

$$x[n + N] = x[n] \quad \forall n, \quad N \in \mathbb{Z}^+$$

## Frequency Condition for Periodicity

For  $x[n] = A \cos(2\pi f_0 n + \phi)$ , the signal is periodic iff

$$f_0 = \frac{K}{N}, \quad K, N \in \mathbb{Z}$$

i.e.,  $f_0$  is a **rational number**.

Rational  $f_0 \Rightarrow$  Periodic

Example:  $f_0 = \frac{1}{4}$ , period  $N = 4$

Irrational  $f_0 \Rightarrow$  Aperiodic

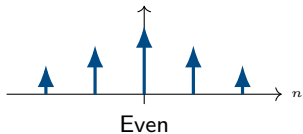
Examples:  $f_0 = 3.33\bar{2}$ ,  $1.71\bar{4}$

# Even and Odd Signals

## Even Signal

$$x[n] = x[-n]$$

Symmetric about  $n = 0$ .

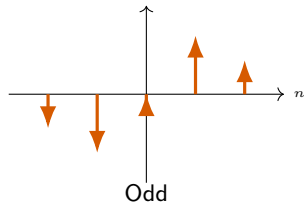


## Odd Signal

$$x[n] = -x[-n]$$

Anti-symmetric about  $n = 0$ .

Note:  $x[0] = 0$  always.



# Even–Odd Decomposition

## Theorem

*“Any discrete-time signal can be expressed as the summation of **one even** and **one odd** component.”*

$$x[n] = x_e[n] + x_o[n]$$

# Even–Odd Decomposition

## Theorem

*“Any discrete-time signal can be expressed as the summation of **one even** and **one odd** component.”*

$$x[n] = x_e[n] + x_o[n]$$

## Derivation:

$$x[n] = x_e[n] + x_o[n] \tag{1}$$

$$x[-n] = x_e[n] - x_o[n] \tag{2 (substitute  $-n$ )}$$

# Even–Odd Decomposition

## Theorem

*“Any discrete-time signal can be expressed as the summation of **one even** and **one odd** component.”*

$$x[n] = x_e[n] + x_o[n]$$

## Derivation:

$$x[n] = x_e[n] + x_o[n] \tag{1}$$

$$x[-n] = x_e[n] - x_o[n] \tag{2 (substitute  $-n$ )}$$

Adding and subtracting (1) and (2):

$$x_e[n] = \frac{1}{2} [x[n] + x[-n]] \quad x_o[n] = \frac{1}{2} [x[n] - x[-n]]$$



# Even–Odd Decomposition: Problem Statement

## Example

Decompose the discrete-time signal

$$x[n] = \{1, 2, 3, 4\}$$

where

$$n = 0, 1, 2, 3$$

into:

$$x_e[n] \quad \text{and} \quad x_o[n]$$

## Step 1: Find the Folded Signal $x[-n]$

Original signal:

$$x[n] = \{1, 2, 3, 4\}$$

with indices:

$$n = 0, 1, 2, 3$$

### **Time Reversal (Folding)**

Flip the signal around  $n = 0$ :

$$x[-n] = \{4, 3, 2, 1\}$$

with indices:

$$n = -3, -2, -1, 0$$

## Step 2: Compute the Even Part

Formula:

$$x_e[n] = \frac{1}{2} (x[n] + x[-n])$$

### Example Calculations

$$x_e[0] = \frac{x[0] + x[0]}{2} = \frac{1 + 1}{2} = 1$$

$$x_e[1] = \frac{x[1] + x[-1]}{2} = \frac{2 + 0}{2} = 1$$

$$x_e[-1] = \frac{x[-1] + x[1]}{2} = \frac{0 + 2}{2} = 1$$

## Step 3: Compute the Odd Part

Formula:

$$x_o[n] = \frac{1}{2} (x[n] - x[-n])$$

### Example Calculations

$$x_o[0] = \frac{1 - 1}{2} = 0$$

$$x_o[1] = \frac{2 - 0}{2} = 1$$

$$x_o[-1] = \frac{0 - 2}{2} = -1$$

**Observation:**

$$x_o[n] = -x_o[-n]$$

So, the odd part is antisymmetric.